

HW 5

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This homework is meant to give you practice in creating and defending a position with both statistical and philosophical evidence. We have now extensively talked about the COMPAS ¹ data set, the flaws in applying it but also its potential upside if its shortcomings can be overlooked. We have also spent time in class verbally assessing positions both for and against applying this data set in real life. In no more than two pages ² take the persona of a statistical consultant advising a judge as to whether they should include the results of the COMPAS algorithm in their decision making process for granting parole. First clearly articulate your position (whether the algorithm should be used or not) and then defend said position using both statistical and philosophical evidence. Your paper will be grade both on the merits of its persuasive appeal but also the applicability of the statistical and philosophical evidence cited.

The purpose of parole is to offer individuals the opportunity to reintegrate into society, incentivize good behavior, and help reduce overcrowding in prisons (“Parole”). Whether a prisoner is eligible for parole is typically decided by a legislative board and judge, who consider the individual’s history and time in prison. As a statistical consultant, I advise against using the results of the COMPAS algorithm in parole decisions. The algorithm has been shown to exhibit racial bias and fails to meet both statistical and ethical standards for fair decision-making. Its inclusion in the parole process undermines the fairness of the system and the moral goals of parole, making its use inappropriate and unreliable.

Statistically, the COMPAS algorithm has limited predictive power, with an accuracy rate of less than 70%. While this may be acceptable in some contexts, a decision as impactful as granting parole requires a higher level of accuracy. Though COMPAS shows comparable accuracy across racial groups, we must also assess bias using statistical measures of fairness to understand the full scope of its impact. One way to measure bias is through disparate impact, which examines whether a model’s predictions are independent of a protected class, in this case, race. This is done by comparing the proportion of individuals from different racial groups who are predicted to be high risk. If the ratio of predicted high-risk individuals between Black and White groups falls below the typical threshold of 0.8, the model is said to exhibit disparate impact. In the case of COMPAS, this ratio is 0.5916, which is below 0.8 This means the algorithm disproportionately categorizes Black individuals as high-risk, leading to unfair outcomes based on race. It is important to note that disparate impact alone does not account for the accuracy of the algorithm’s predictions, meaning it does not assess whether the individuals predicted to be high-risk are actually high-risk. To assess accuracy, we consider equalized odds, which measures whether the algorithm’s misclassifications are evenly distributed between groups. Specifically, we compare the rates at which Black and White individuals are correctly and incorrectly classified as high-risk. In the case of COMPAS, the difference between Black individuals misclassified as high-risk and White individuals misclassified as high-risk is 0.218, which is greater than the typical error of 0.2. This suggests that COMPAS misclassifies Black individuals as high-risk more often than White individuals, indicating bias in its predictions. These statistical measures show that COMPAS is not a fair or reliable tool for parole decisions. Its use would lead to racial disparities in parole outcomes, undermining the fairness of the parole system.

In addition to statistical issues, there are significant moral concerns associated with using COMPAS in parole decisions. Misclassifying Black individuals as high-risk more often than White individuals has real-world consequences that disproportionately affect the lives of Black individuals. A key concept of justice

¹<https://www.propublica.org/datastore/dataset/compas-recidivism-risk-score-data-and-analysis>

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is merit, recognizing an individual's growth and rehabilitation during their time in prison. Parole is meant to offer individuals the chance to demonstrate their positive change, yet COMPAS relies on past behaviors, failing to capture the full scope of an individual's growth. People can transform, often in ways that are not reflected in past criminal behavior, and this transformation is sometimes best recognized through human judgment, empathy, and understanding. COMPAS cannot account for this human capacity for change. By relying on an algorithm that overlooks these factors, we undermine the essence of fairness in parole decisions.

From a deontological perspective, judges have an ethical duty to make decisions free from bias. They are sworn to uphold fairness, which includes making decisions based on individual circumstances. Using an algorithm that has been shown to be biased directly violates this duty. Furthermore, COMPAS is a "black box" model, meaning its decision-making process is unknown to us. Judges who rely on COMPAS cannot explain how the algorithm arrived at its decisions, which undermines the transparency and accountability required in the judicial process. This lack of transparency also prevents individuals from understanding the basis for their parole denial, which further decreases trust in the system.

We can also consider the two main deontological formulations, universal maximum and moral agents as a mere means to an end. If we were to universalize the use of COMPAS in parole decisions, we would perpetuate racial bias and unfair treatment in the legal system. It would institutionalize racial disparities in the criminal justice system, worsening societal tensions and continuing to unjustly affect Black individuals disproportionately. This would contradict the principles of justice and equality while further oppressing members of the Black community. Additionally, under the 'means to an end' formulation of deontology, COMPAS could be seen as using past inmates as a mere means to an end. The algorithm is trained on historical data. This data is then used to predict future behavior. In this way, the past experiences of these individuals are being used to shape decisions that impact others' lives, without regard for their autonomy. This creates a self-reinforcing cycle of bias, as the algorithm perpetuates the biases present in the past data.

Ultimately, the purpose of parole is to offer individuals a chance to demonstrate their growth, rehabilitation, and potential to reintegrate into society. By using COMPAS, judges would be relying on an algorithm that is statistically unreliable and philosophically flawed. This would limit individuals' opportunities for fair consideration and perpetuate racial disparities in parole decisions. Because COMPAS fails to meet the necessary standards of fairness and justice, both statistically and ethically, I strongly advise against incorporating its results into parole decisions.

Works Cited

“Parole.” LII / Legal Information Institute, Cornell Law School, Nov. 2023, www.law.cornell.edu/wex/parole.